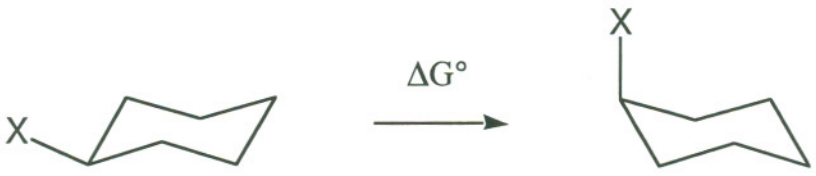
# ORGANIC CUME

September 12, 2008

(1) Give typical values for the bond length (to 0.1Å) and homolytic bond dissociation energies (± 5 kcal/mol) of the following bonds.

| | | |
|---|---|---|
| C—C | C—O | C—N |
| C═C | C═O | C═N |
| C≡C | | C≡N |
| | C—H | |
| | O—H | |

(2) Give the A-values (to 0.1 kcal/mol) for the following groups X.

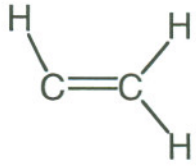



$CH_3$

$OCH_3$

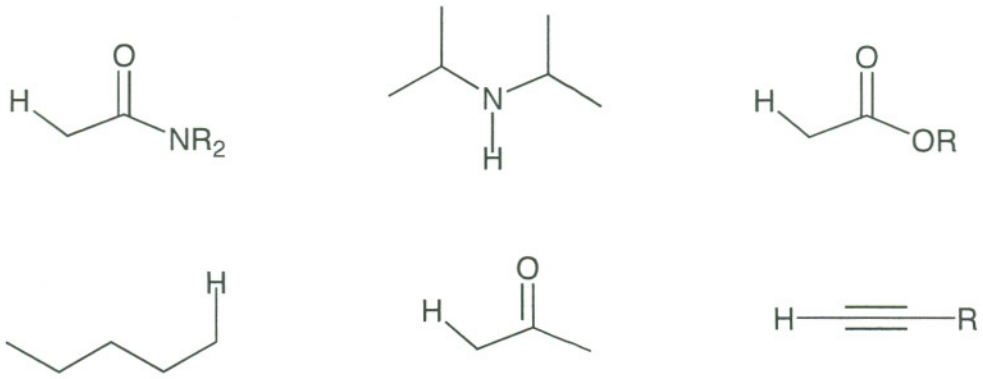
phenyl

t-butyl




C≡C—H

(3) Given two equilibrating species A and B, what energy difference in kcal/mol does there have to be between A and B to afford a 10:1 ratio of A:B at 298 K?

(4) What is the approximate vicinal coupling constant between two protons which possess an H-C-C-H dihedral angle of 180°? Of 0 °?

(5) Rank the acidity from most to least acidic and give the approximate $pK_a$'s of the following compounds.